信号与系统实验

主讲人：吴光 博士

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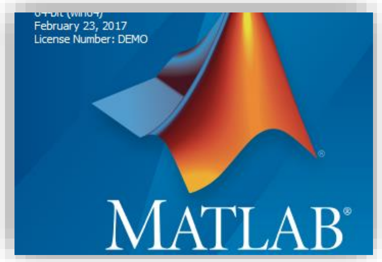
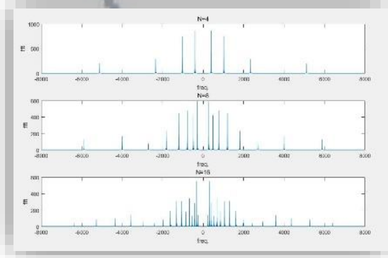
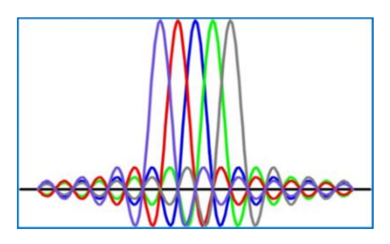
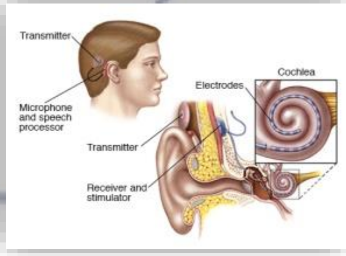








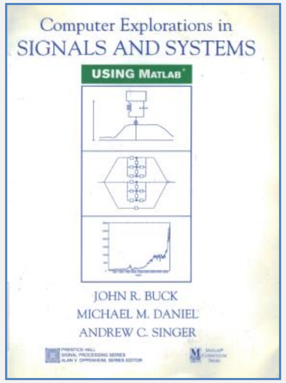
Signals and Systems (Lab)

Project 2: Motion detection via communication signals

Dr. Wu Guang
wug@sustech.edu.cn
Electrical & Electronic Engineering
Southern University of Science and Technology

Labs in this course

5 Lab assignments+2 Projects



- Lab 1: MATLAB Programming (3 Weeks)
- Lab 2: Linear Time-Invariant Systems (2 Weeks)
- Lab 3: Fourier Series Representation of Periodic Signals (2 Weeks)
- Lab 4: The Continuous-Time Fourier Transform (2 Weeks)
- Lab 5: Coding Test & System, Transform, Convolution and Filter (1 Week)
- Project 1: Speech synthesis and perception with envelope cue (3 Weeks)
- **Project 2: Motion detection via communication signals (3 Weeks)**

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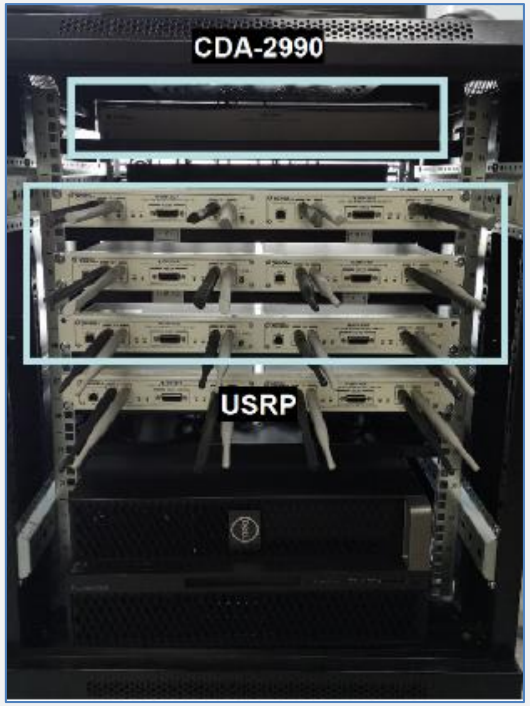
1.1 Research：Passive Motion Detection

ISAC： Integrated Sensing And Communications （通信感知一体化）



多天线采集系统

运动检测实验场景

Passive Motion Detection via mmWave Communication System

Abstract—**In this paper, an integrated passive sensing and communication system working in 60 GHz band is elaborated, and the sensing performance is investigated in an application of hand gesture recognition. Specifically, in this integrated system, there are two radio frequency (RF) chains at the receiver and one at the transmitter. Each RF chain is connected with one phased array for analog beamforming. To facilitate simultaneous sensing and communication, the transmitter delivers one stream of information-bearing signals via two beam lobes, one is aligned with the main signal propagation path and the other is directed to the sensing target. Signals from the two lobes are received by the two RF chains at the receiver, respectively. By cross ambiguity coherent processing, the time-Doppler spectrograms of hand gestures can be obtained. Relying on the passive sensing system, a dataset of received signals, where three types of hand gestures are sensed, is collected by using Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) paths as the reference channel respectively. Then a neural network is trained by the dataset for motion detection. It is shown that the classification accuracy rate is high as long as sufficient sensing time is assured. Finally, an empirical model characterizing the relation between the classification accuracy and sensing duration is derived analytically.**

I. INTRODUCTION

Integrated Sensing and Communication (ISAC) is a promising technology for the sixth generation cellular system, where the sensing capability has potential to assist the wireless communications and provide new services to subscribers [1], [2]. Compared with sensing of vehicle's velocity and position, human motion sensing is usually more challenging, as the micro-Doppler effect should be captured for classification. Generally, there are three main approaches of wireless human motion sensing in existing ISAC testbed implementations, including wireless sensing via dedicated waves (e.g., Frequency-Modulated Continuous Wave, FMCW), wireless sensing via channel state information (CSI), and passive sensing.

There have been a significant number of works using radar for human motion recognition. For example, in [3], the authors proposed the first coarse multi-person gesture tracking system with FMCW radar, where the direction of a pointing hand can be identified. In [4], Google developed a commercial FMCW-based gesture recognition system, namely the Soli project. In addition, it was shown in [5] that the motion of human skeleton could be reconstructed with recent advances in deep learning.

Compared with integrated radar and communication systems where dedicated time or frequency resources should be reserved for sensing, there are a number of research works detecting human motion via channel state information (CSI), where the channel sensing does not raise overhead on wireless resource. Particularly, the micro-Doppler effect can also be observed from the variation of the CSI. Moreover, the location of sensing target can be estimated via CSI if multi-antennas and multi-carrier technologies are adopted in the communication system. For example, the CSI was exploited to detect human motions behind the wall in [6]. In [7], the authors explored CSI histograms to recognize daily activities, such as cooking in a kitchen and walking from one room to another, based on empirical study. In [8], the authors proposed theoretical models to derive the relation between CSI and human activities. Recent work [9] jointly estimated the Angle-of-Arrival (AoA), Time-of-Flight (ToF), and Doppler-Frequency-Shift (DFS) for human localization and tracking.

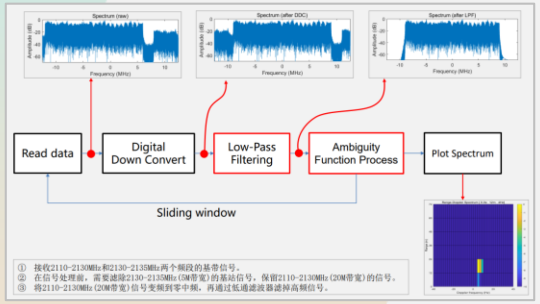
Despite low overhead, CSI-based sensing methods are sensitive to the frequency offset between the transmitter and receiver, which might be a severe problem in millimeter wave (mmWave) band. Although there are methods to eliminate the issue of frequency offset [8], the performance relies on the propagation path without Doppler shift. On the other hand, passive radar is a sensing approach insensitive to the frequency offset. In [10], passive sensing via WiFi signals was adopted to detect the moving personnel. It was further shown in [11] that human breathing behind the wall could be detected via passive sensing. Moreover, the CSI-based sensing and passive sensing were compared in [12]. It was shown that CSI-based system performed better in Line-of-Sight (LoS) scenarios, while passive radar system performed better in Non-Line-of-Sight (NLoS) scenarios.

Although there have been a number of testbeds and experiment results on the passive sensing in sub-6 GHz band, there is still no passive sensing testbed in mmWave band. Note that communications in mmWave band has already been considered in the 5G cellular system and IEEE 802.11ay systems, it is natural to extend the passive sensing technique to mmWave band. Moreover, because of the smaller wavelength, sensing in mmWave band would have higher Doppler resolution.

In this paper, 60 GHz mmWave communication and passive sensing system implementation is elaborated. The 16-antenna phased arrays are deployed at both the transmitter and receiver, so that the mmWave signal transmission can be directed to the sensing target to suppress the interference from LoS path. The performance of the above system is investigated via the exemplary application of hand gesture recognition. Specifically, a dataset of received signals of three gestures is provided and

“Passive Motion Detection via mmWave Communication System,” 2022 IEEE 95th Vehicular Technology Conference: (VTC2022), Helsinki, Finland, 2022.

1.2 Teaching Process



- 了解实验目标
- 激发实验兴趣
- 融入思政内容

- 多径效应
- 多普勒频移
- 参考信道模型
- 监视信道模型
- 模糊函数
- 数字下变频
- 数字低通滤波器
- 距离/时间-多普勒谱

Human Motion Detection

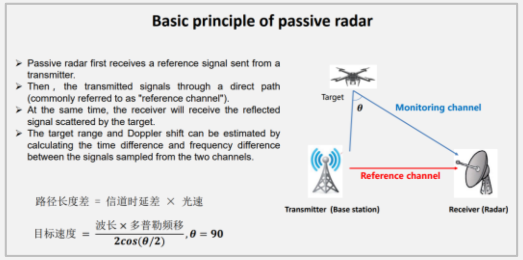
Output Waveform

Project Start → **Background** → **Concepts** → **Outdoor Experiment** → **MATLAB Programming** → **Presentation** → Report

Aircraft Detection

Principle

On-Line/Off-Line




NOCFEE
证 书
为表彰"2023年第十届全国高校电工电子基础课程实验教学案例设计竞赛（鼎阳杯）"获奖者，特颁发此证书。
作品名称：基于通信信号的运动检测
获 奖 者：吴光、王小静、王锐
学　　校：南方科技大学
获奖等级：中南赛区二等奖
中国电子学会
高等学校国家级实验教学示范中心联席会电子学科组
2023年5月

SUSTech Southern University of Science and Technology

1.3 From Principle to Ability

Focus： I have an ability to ......

解决复杂工程问题的能力

Complex Problem: How to use communication signals to detect moving objects ?

研究

低通滤波器　数字下变频　信号的频谱　模糊函数

编程能力　演讲能力

写作能力　团队沟通

$$|H(\omega)| = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_c}\right)^{2n}}}$$

$$y_{\mathrm{ddc}}(n) = y(n)e^{-j2\pi f_0 n}$$

$$Y_{\mathrm{ddc}}\left(e^{j\omega}\right) = Y\left(e^{j2\pi(f+f_0)n}\right)$$

$$Cor(\tau, f_{\mathrm{d}}) = \int_t^{t+T} y_{sur}(t-\tau)y_{ref}^*(t)e^{-j2\pi f_{\mathrm{d}}t}\,dt$$

原理

线性系统　卷积性质　傅里叶变换　信号互相关

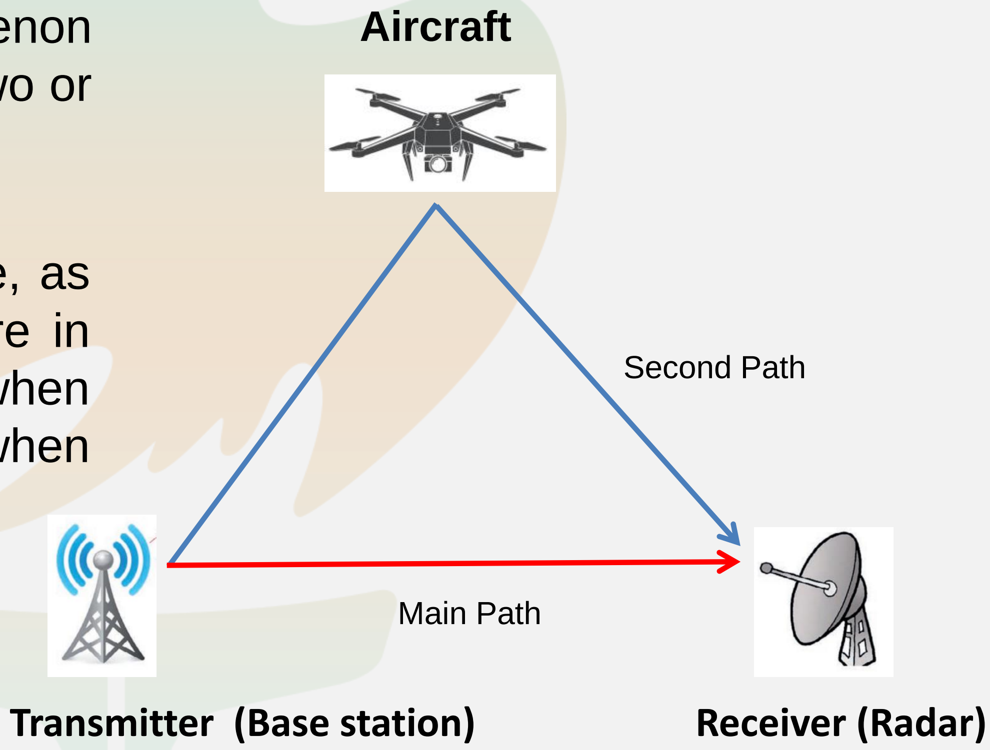
复杂性

2.1 Concept: Multipath propagation

Multipath propagation: Multipath is the propagation phenomenon that results in radio signals reaching the receiving antenna by two or more paths. (Wikipedia)

Doppler shift: A change in the observed frequency of a wave, as of sound or light, occurring when the source and observer are in motion relative to each other, with the frequency increasing when the source and observer approach each other and decreasing when they move apart. (Wikipedia)

Radar : A method of detecting distant objects and determining their position, velocity, or other characteristics by analysis of very high frequency radio waves reflected from their surfaces. (Wikipedia)



Multipath effect

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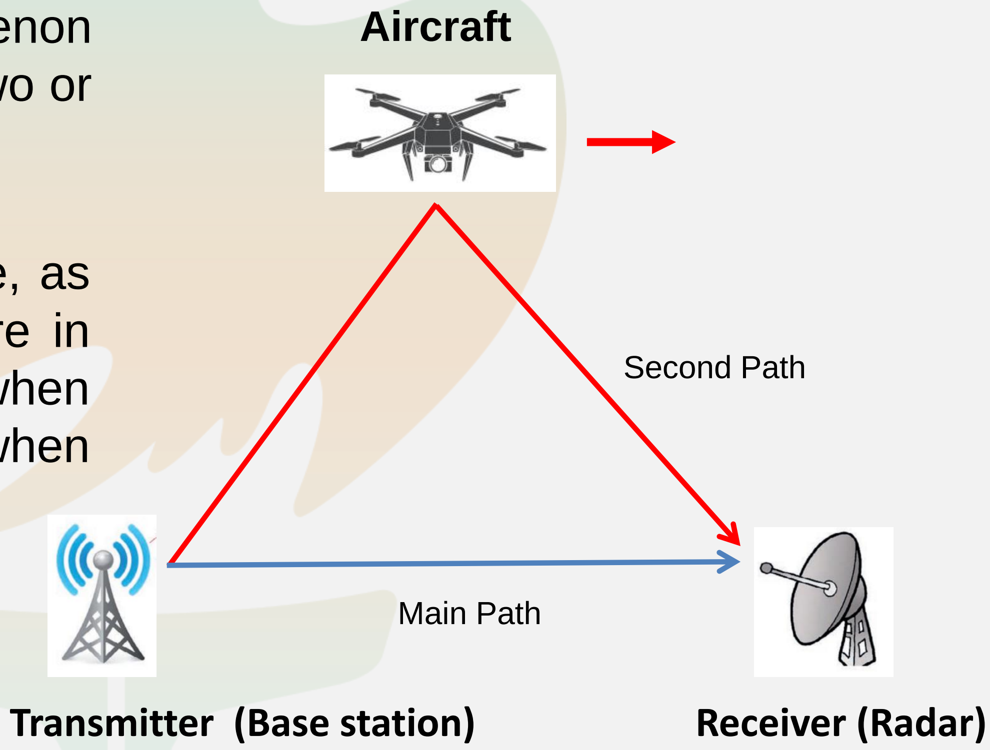
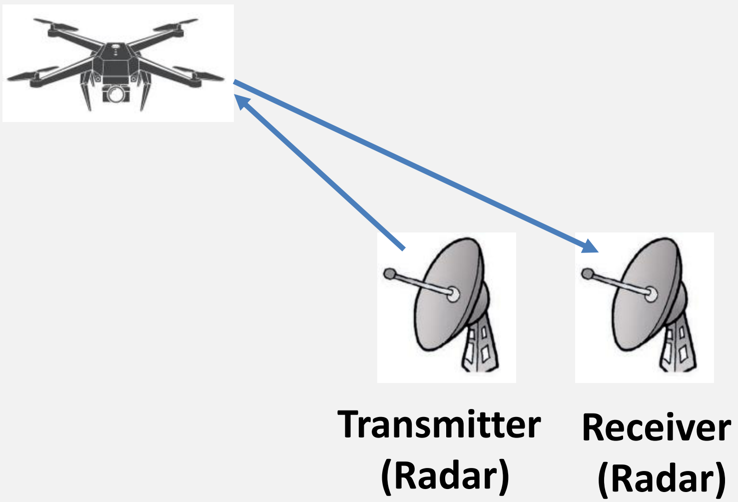
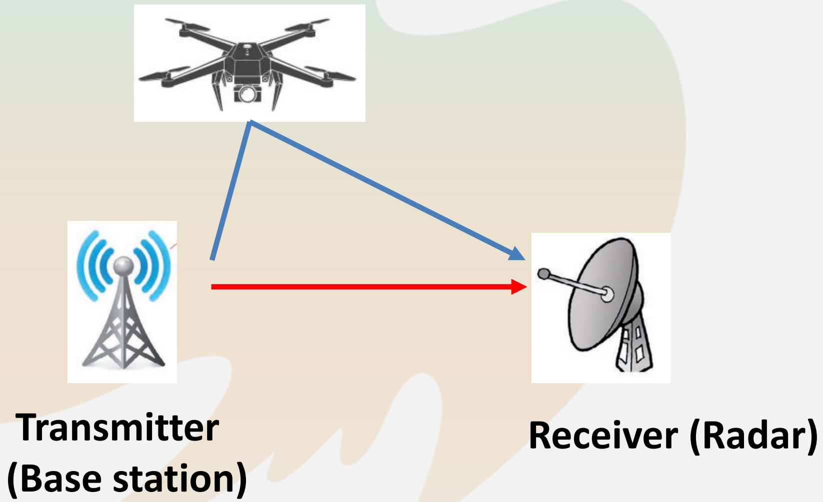


Multipath effect

3. Concept：Active radar or Passive radar



Active radar



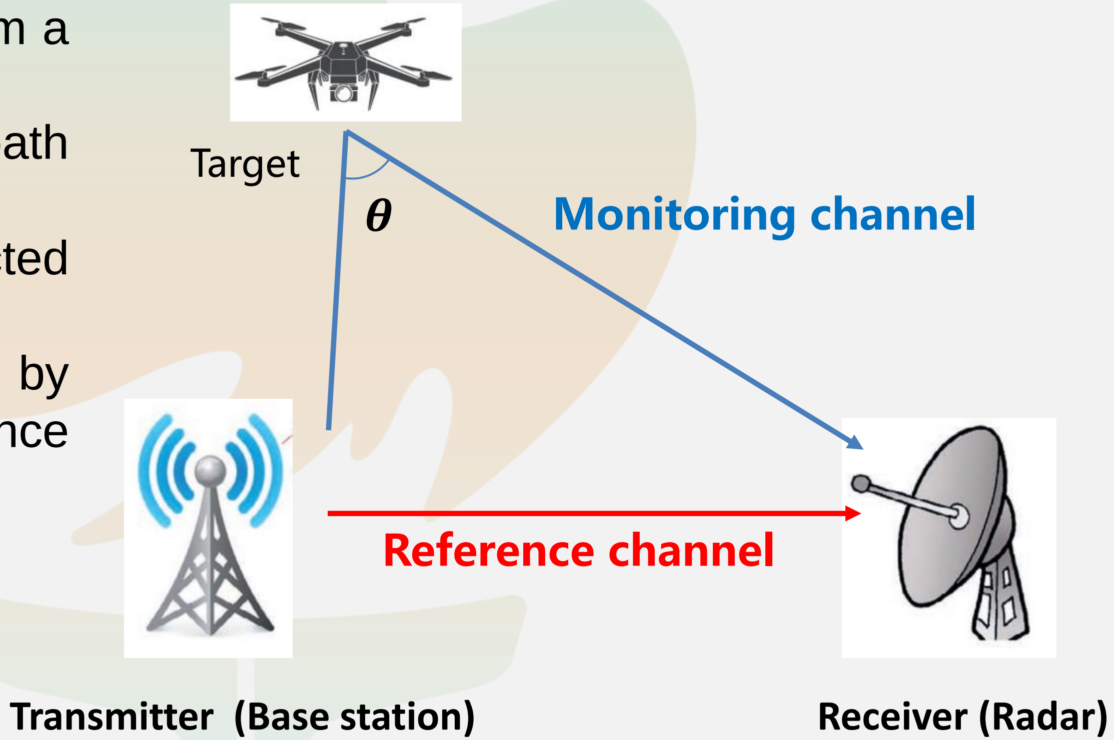
Passive radar

In military applications,

- ➢ Passive radar keeps radio silence during target detection, which is difficult to locate or be interfered, so it is very suitable for scenes with high requirements for concealment.
- ➢ No additional time / frequency resources are required, which makes its cost and complexity much lower than that of traditional active radar equipment.

4.1 Basic principle of passive radar

- Passive radar first receives a reference signal sent from a transmitter.
- Then，the transmitted signals through a direct path (commonly referred to as "reference channel").
- At the same time, the receiver will receive the reflected signal scattered by the target.
- The target range and Doppler shift can be estimated by calculating the time difference and frequency difference between the signals sampled from the two channels.



路径长度差 = 信道时延差 × 光速

$$\text{目标速度} = \frac{\text{波长} \times \text{多普勒频移}}{2cos(\theta/2)}, \theta = 90$$

4.2 Basic principle of passive radar

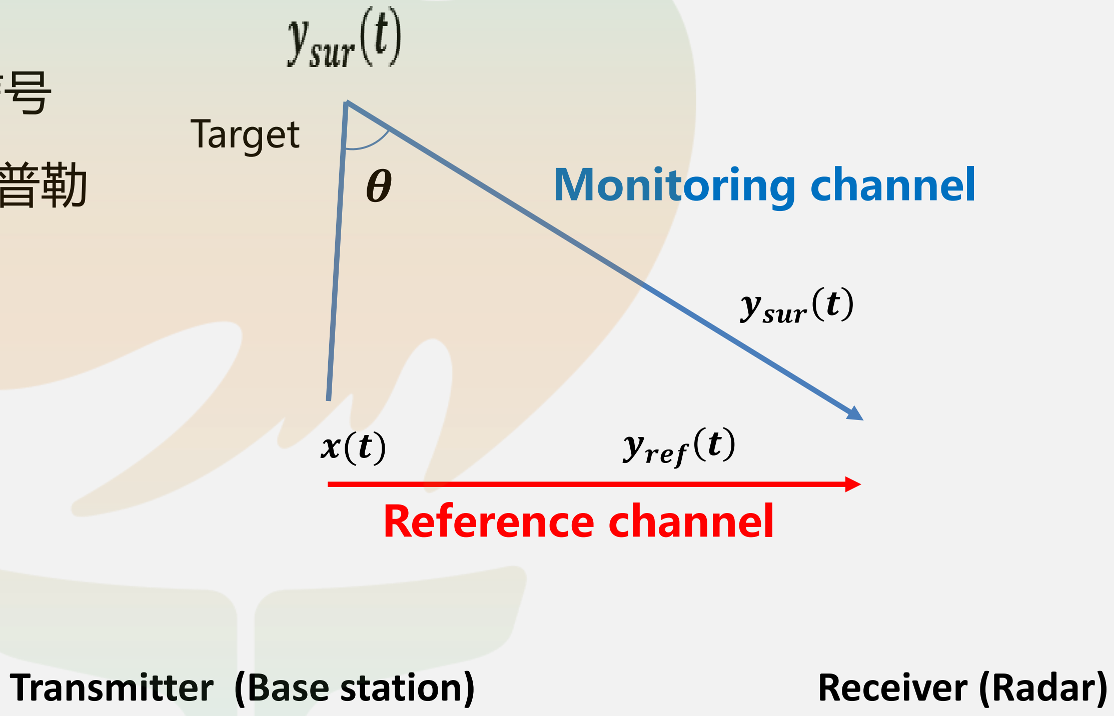
- 假设 $\boldsymbol{x(t)}$ 为发射信号，
- 参考信号$\boldsymbol{y_{ref}(t) = \alpha x(t-\tau_r)}$ 为时延和衰减后的发射信号
- 监视信号$\boldsymbol{y_{sur}(t) = \beta x(t-\tau_s) e^{j2\pi f t}}$ 为衰减、时延和多普勒频偏后的信号。

- 定义模糊函数，

$$Cor(c,d) = \int_t^{t+T} y_{sur}(t+c) y_{ref}^*(t) e^{-j2\pi d t}\, dt$$

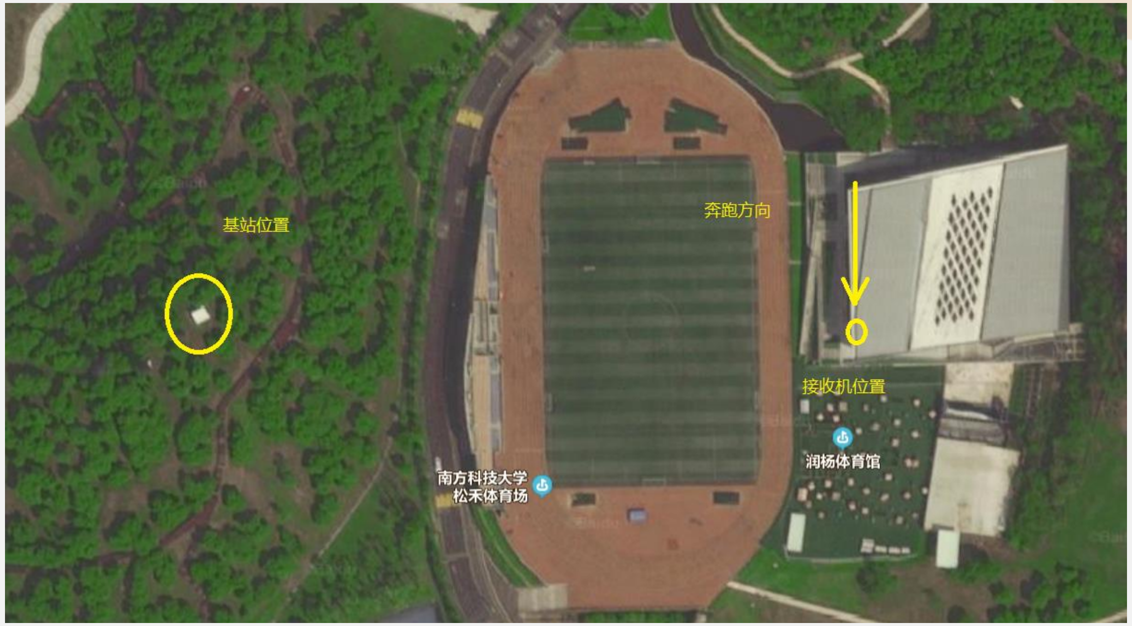
- 估计时延差和多普勒频移 (τ, f):

$$(\hat{\tau}, \hat{f}) = \arg\max_{c,d} Cor(c,d)$$

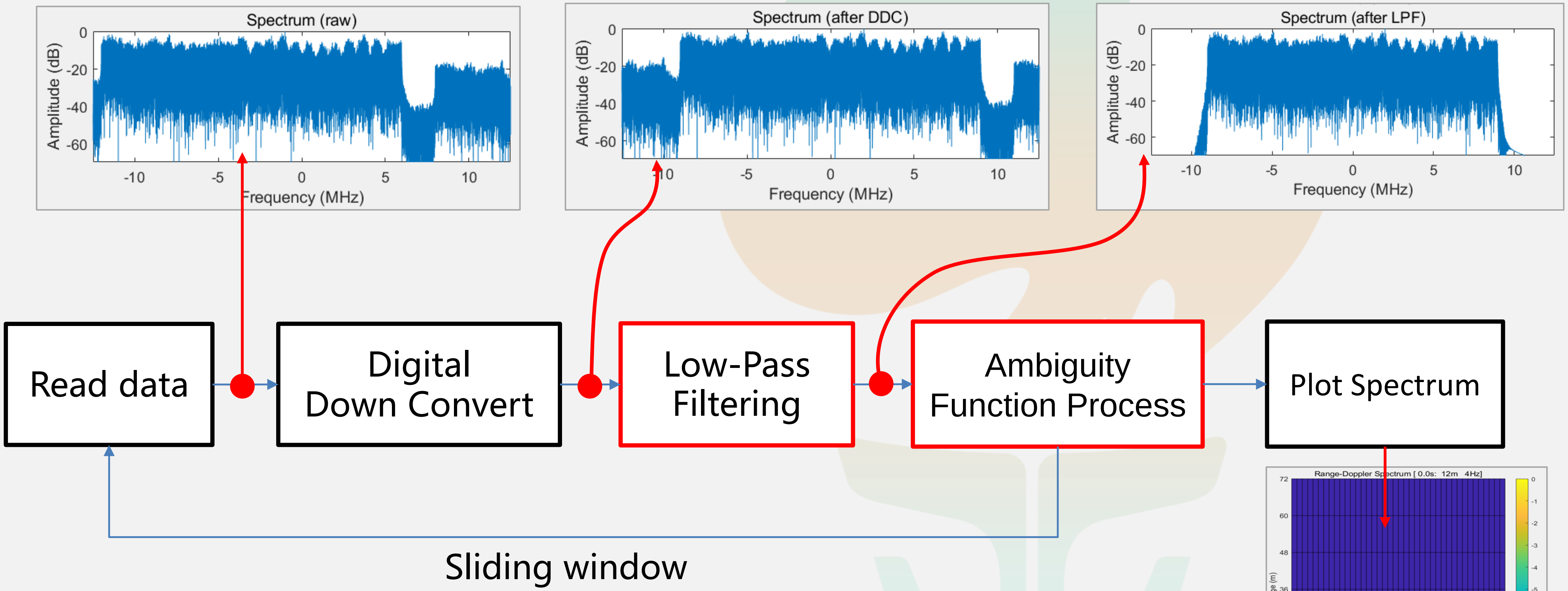


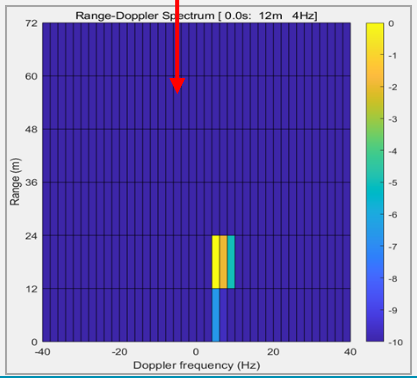
5. Outdoor Experiment

- 实验地点在松和体育场。一个8天线单元的阵列天线对着远处基站，另一个8天线单元的阵列天线对着一名学生，该学生在天线前来回跑动，每台USRP-2954R可以接收两个通道的射频信号，并将射频信号下变频至基带，USRP-2954R通过10G光网口将基带信号上传至计算机中存储。6台软件无线电平台USRP-2954R连接到同一时钟分配器NI CDA-2990。整个实验场景如图所示。



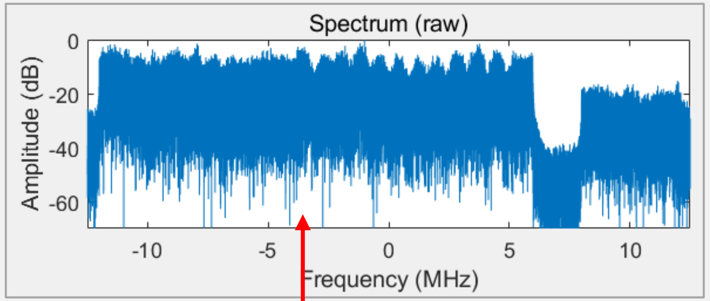
6.1 Digital Signal Process

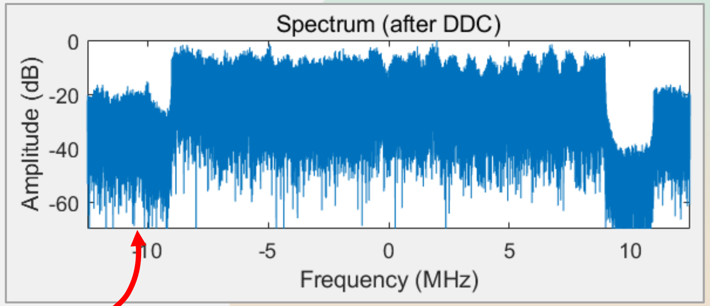


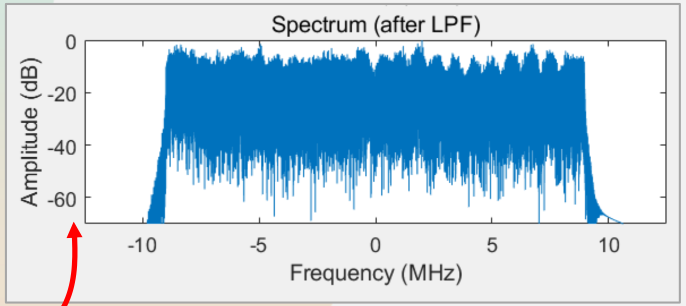


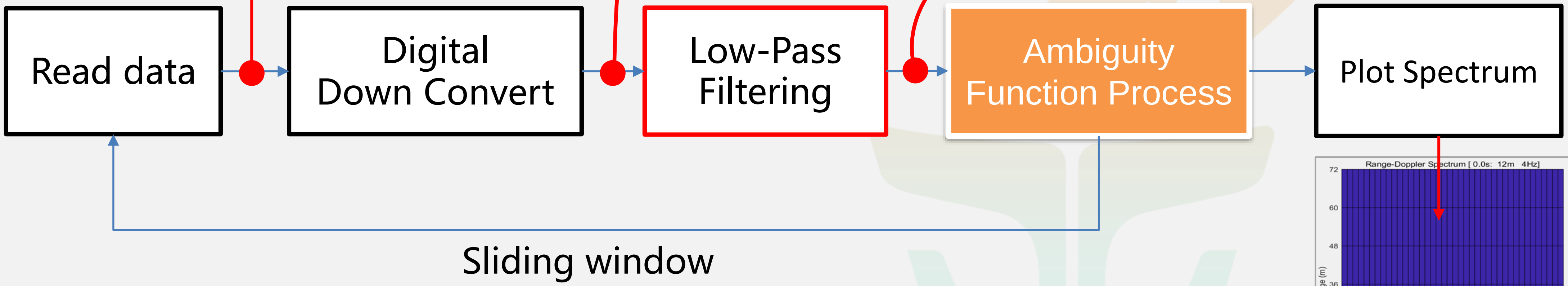
① 接收2110-2130MHz和2130-2135MHz两个频段的基带信号。
② 在信号处理前，需要滤除2130-2135MHz(5M带宽)的基站信号，保留2110-2130MHz(20M带宽)的信号。
③ 将2110-2130MHz(20M带宽)信号变频到零中频，再通过低通滤波器滤掉高频信号。

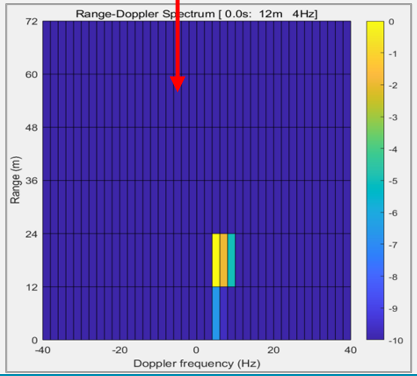
6.2 Ambiguity Function Process











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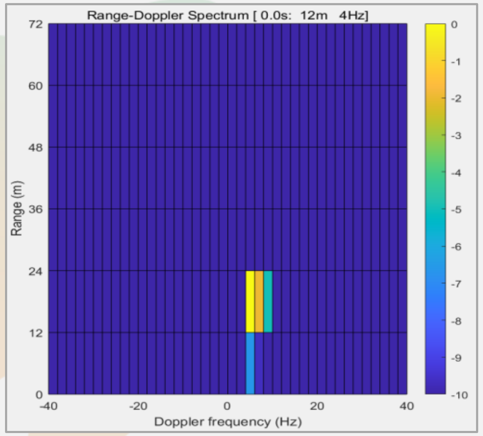
7. Time-Doppler Spectrum

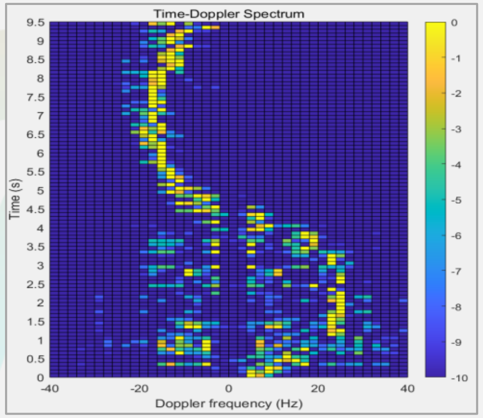
- 在两路接受信号中取$[t, t+T]$时间内的片段进行模糊函数计算，通过遍历所有可能的(c, d)求模糊函数值，选取最大值对应的(c, d)作为估计值。



$$Cor(c, d, \mathrm{t}) = \int_{t}^{t+T} y_{\mathrm{sur}}(t+c)\, y_{\mathrm{ref}}^{*}(t)\, e^{-j2\pi dt}\, dt$$

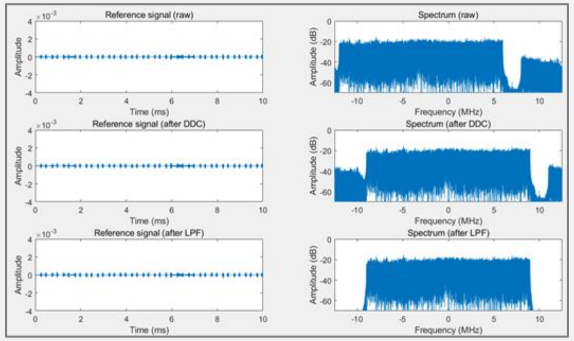
$$g(d, t) = \max_{c} |Cor(c, d, t)|$$

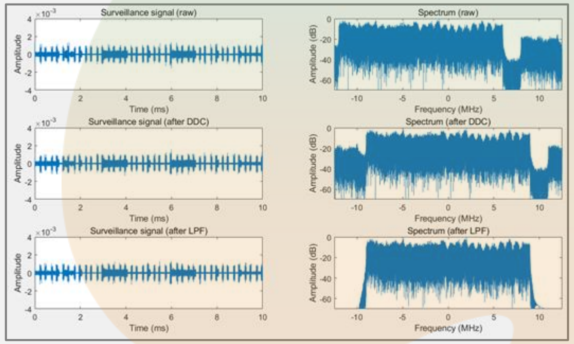
- 将$g(d, t)$向量化为$\hat{g}(t)$，绘制出$\hat{g}(t)$随时间的变化关系



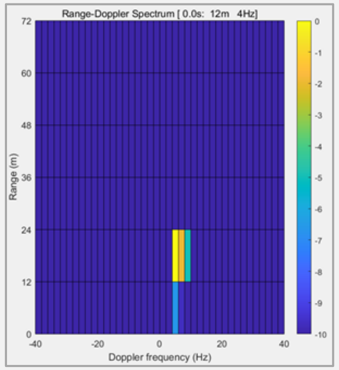
8. Project Tasks

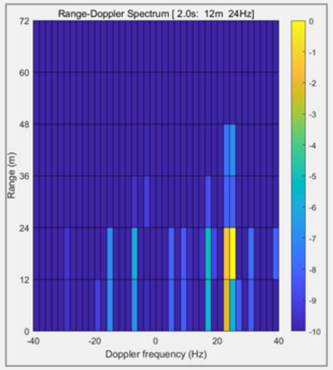
（1）参考信号和监测信号的原始、变频后、低通滤波后的信号时域波形和频谱：

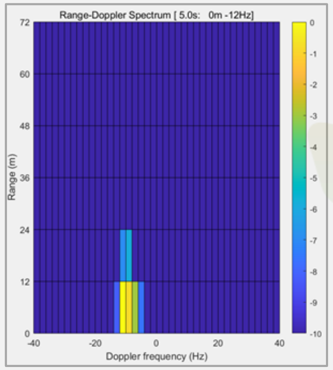


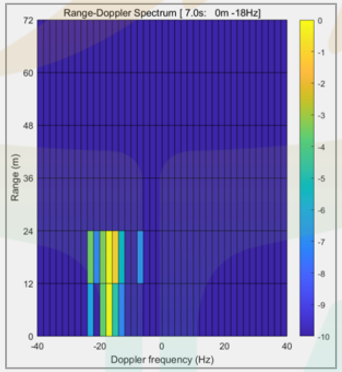


（2）0-0.5s、2-2.5s、5-5.5s、7-7.5s信号处理后得到的距离-多普勒谱：

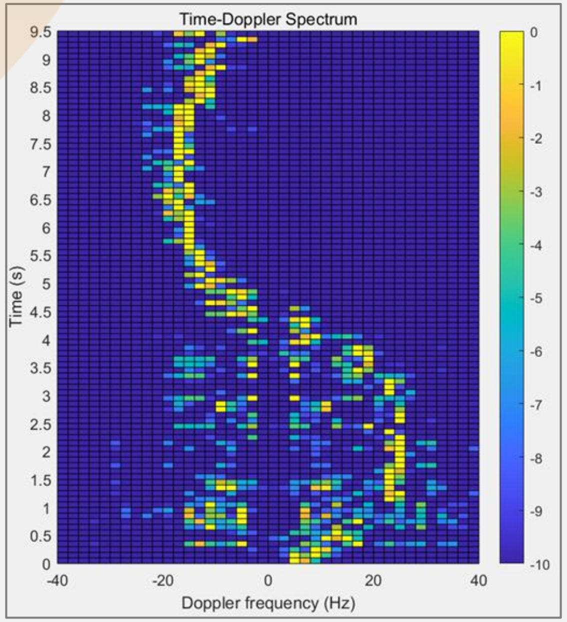








（3）以CIT=0.5s为滑动窗口每次滑动0.5s处理得到的10s数据的多普勒-时间谱：



8. Organization

- **Each group consists of four students.**
- **Each group need present one Lab project (submit reports for both projects):**
  - The presentation date
  - **Each presentation is 10 minutes (including Q & A)**
  - <u>All team members need to contribute to the presentation.</u>
  - Presentation in English (recommended) or Chinese.

Question ?